Supplementary Material Outline: BodyFutures-Bench

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1 Environment Specifications

The supplement will provide the complete equations and locked configuration values for each BodyFutures environment.

Table 1: Supplement environment equation checklist.

Environment	Required details
FeverAnt	Temperature, energy, torque damage, actuator weakening, FAC, collapse thresholds, observation modes.
WoundWalker	Wound accumulation, impact load, balance health, energy, actuator weakening, FAC, collapse thresholds.
ThermalHopper	Heat from action and landing impact, cooling, damage, energy, actuator weakening, FAC.
NumbArm	Numbness, proprioceptive corruption, manipulation load, damage, energy, action rescaling, success criterion, FAC.

2 Preregistered Protocol

This section will reproduce the frozen CoRL manifest: seeds, train budgets, episode lengths, evaluation horizons, baseline list, checkpoint-selection score, IWM dataset mixture, coverage gates, and final metric definitions. Any deviation from the manifest will be listed with a version identifier and reason.

3 Run Commands

The supplement will include command-line recipes for reproducing the final experiments:

```
bash experiments/run_corl_2026_rescue_primary.sh
```

```
bash experiments/run_corl_2026_primary.sh
bash experiments/run_corl_2026_secondary.sh
bash experiments/run_corl_2026_all.sh
```

For archival final sweeps, the exact manifest version, output root, run id, and seed set will be reported.

4 Weights and Biases Metric Schema

All scalar metrics will be logged against a shared global step. The supplement will list stable metric keys for training, rollout diagnostics, body state, evaluation, inference, IWM training, and policy video. Step-specific video filenames may exist locally for traceability, but W&B keys will remain stable.

5 IWM Dataset Coverage

For each environment and seed, the supplement will report:

- transition count and episode count by collection mode;
- FAC minimum, mean, standard deviation, and histogram;
- fraction of samples below FAC thresholds 0.75 and 0.50;
- collapse-adjacent fraction and collapse labels;
- whether automatic coverage repair was invoked.

6 IWM Architecture and Training

The supplement will provide model dimensions, ensemble size, input layout, horizon heads, loss weights, optimizer settings, batch size, device placement, and calibration diagnostics. For final runs, ensemble size and training settings will be frozen before seed sweeps.

7 Evaluation Details

The supplement will define:

- task reward per step and cumulative task return;
- survival fraction and collapse rate;
- final FAC, minimum FAC, viability cost, and body score;
- early-warning time before collapse;
- prediction RMSE, low-FAC error, calibration, AUROC, and AUPRC;
- bootstrap or Student confidence interval computation.

8 Compute Budget

The final supplement will report CPU model, GPU model, RAM, number of vector environments, wall-clock runtime per method, IWM training time, and total compute used for final runs. Pilot and debugging compute will be reported separately from frozen result compute.

9 Video Storyboard

The video supplement will show:

1. FeverAnt task PPO sprinting early and degrading over long horizons;
2. FeverAnt PPO-IWM preserving FAC while retaining task progress;
3. WoundWalker comparison under long-horizon degradation;
4. IWM calibration visualization and FAC trajectories;
5. representative failure cases where IWM is overconservative or miscalibrated.

10 Failure Cases and Negative Results

The supplement will explicitly include runs where PPO-IWM fails to improve over current-viability PPO or loses too much task reward. These cases are important for identifying whether the contribution is stronger as a benchmark and prediction protocol, or as an algorithm that reliably improves the full task-viability Pareto frontier.